Harsh Sharma
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Education:-

- **BTech Major :- Robotics & Artificial Intelligence (Year 2024-2028) from KJSSE**

About Me:-

B.Tech undergraduate in Robotics & Artificial Intelligence with a growing focus on **humanoid robotics** — specifically real-time perception, motion planning, and embedded control for autonomous physical systems. I've built hands-on experience in computer vision (stereo vision, object detection/segmentation with YOLO), ROS2-based robot communication, and closed-loop hardware control through my work as **Image Processing & MATLAB Joint Head at Team KJSCE Robocon**, where I design vision-based sensing pipelines and mentor junior engineers on autonomous navigation systems. I'm actively deepening my expertise in deep learning, simulation (Gazebo/Simulink), and embedded systems (ROS2, microcontroller-level sensor/actuator interfacing) — the core building blocks of humanoid perception-to-action pipelines — with the long-term goal of contributing to legged locomotion, manipulation, or human-robot interaction systems.

Experience:-

- **RabTech Solution:- CV Intern for autonomous drones | Aug 2026 - Present | Hybrid –**
- **Fostride:- AI/ML Intern | Apr 2026 – Aug 2026 | Maharashtra, India | On-site | Part-Time –** contributing to smart waste management solutions through robotics, AI, embedded systems, and sustainable technology innovation.
- **Team KJSSE Robocon:- MATLAB & Image Processing Joint Head | Aug 2025 – Aug 2026 | Maharashtra, India | On-site -** Learning and developing various algorithms for autonomous robot navigation using computer vision, concepts of stereo vision, ROS2 communication & hardware control, Tested yolo11 and yolo26 models, Team management and leadership.
- **Team KJSSE Robocon:- MATLAB & Image Processing – Junior Engineer | Sep 2024 – Aug 2025 (1 year) :-** Learnt and developed skills in image processing, computer vision, MATLAB, basics of ROS and ROS2, basic ML algorithms like linear regression

Skills:-

- **Programming languages:-** Python, C, C++, MATLAB
- **Computer Vision:-** OpenCV, Image processing algorithms, Object Detection/Classification/Segmentation, Computer vision based algorithms
- **Embedded systems:-** Arduino, ESP32, ROS2, Communication protocols(Serial, I2C, SPI)
- **Electronics:-** Basic circuit connections, Breadboard connections, Interfacing sensors & actuations with microcontrollers
- **Simulations:-** Simulink, Gazebo
- **AI & Machine Learning:-** Dataset management/manipulation. Model training, YOLO, RFDETR, Neural networks
- **Design & CAD:-** AutoCAD, SOLIDWORK

Projects:- All projects uploaded to github - **github.com/harshsharma356**

- **AgriTech hackathon 2026:-** Automatic pesticide sprinkling system
- **Robocon 2025 Workshop:-** Hand-gesture controlled car using computer vision and esp32
- **Robocon 2025:-** Developed a logic for robot alignment to it's target using hsv thresholding

Licenses & certifications:-

- **Matlab Onramp from Mathworks:-**

https://matlabacademy.mathworks.com/progress/share/certificate.html?id=7723a022-34d2-4a2d-bb08-c18bd7ebfc30&

- **ROS 2 for Beginners (ROS Jazzy - 2026) from Udemy:-**
  https://www.udemy.com/certificate/UC-565f2363-ddca-419d-bb41-0d2ab77a1c3a/